Unit no. ° 8

ALGEBRAIC CALCULATION, EQUATIONS AND INEQUALITIES

4-20 January 2027

I Brackets and distributivity

Removing brackets

After a + sign, removing the brackets keeps the signs of the terms. After the sign $-$, each term inside changes sign. This rule concerns a sum or difference of terms.

Example 8.1

Remove the brackets and simplify: $A = x + (2 + 3x)$; $B = x - (2 + 3x)$; $C = -2x + (-4 + 5x)$; $D = -2x - (-4 + 5x)$.

Solution:

$A = 4x + 2$; $B = -2x - 2$; $C = 3x - 4$; $D = -7x + 4$.

Distributivity

For all real numbers a, b, c, d :

$$b(c + d) = bc + bd, \qquad (a + b)(c + d) = ac + ad + bc + bd.$$

Example 8.2

Expand and simplify:

$$A = -4(8x + 7) + (-8x + 5)(5x + 9),$$
$$B = (6x + 9)(-x + 7) + (3x + 5)(-9x + 6),$$
$$C = -9(-x - 7) - (-x + 3)(-8x - 8),$$
$$D = -5(-6x - 7) + (x - 7)(5x + 8).$$

Solution:

We expand each product, then group terms of the same degree:

$$A = -32x - 28 - 40x^2 - 72x + 25x + 45 = -40x^2 - 79x + 17,$$
$$B = -6x^2 + 33x + 63 - 27x^2 - 27x + 30 = -33x^2 + 6x + 93,$$
$$C = 9x + 63 - (8x^2 - 16x - 24) = -8x^2 + 25x + 87,$$
$$D = 30x + 35 + 5x^2 - 27x - 56 = 5x^2 + 3x - 21.$$

II Factorising with a common factor

Property

$ka+kb=k(a+b)$ and $ka-kb=k(a-b)$. Factorising turns a sum into a product; expanding does the inverse operation.

Example 8.3

Factorise:

$$A = 15x^2 - 21x,$$
$$B = 3(2+3x) - (5+2x)(2+3x),$$
$$C = (4x-1)(x+6) + (4x-1),$$
$$D = (1-6x)^2 - (1-6x)(2+5x).$$

Solution:

$A = 3x(5x-7)$.
$B = (2+3x)[3-(5+2x)] = (2+3x)(-2-2x) = -2(2+3x)(1+x)$.
$C = (4x-1)[(x+6)+1] = (4x-1)(x+7)$.
$D = (1-6x)[(1-6x)-(2+5x)] = (1-6x)(-1-11x) = (6x-1)(11x+1)$. Check the factors by re-expanding.

Example 8.4

Choose a useful form of $f(x) = (x-2)(x+3)$ to calculate $f(2)$, $f(0)$ and find its zeros.

Solution:

The factorised form immediately gives $f(2) = 0$ and the zeros 2, -3. At 0, $f(0) = (-2) \times 3 = -6$; the expanded form $x^2 + x - 6$ also gives this result. The choice depends on the question.

III Special products: expanding

Identities

$$(a+b)^2 = a^2 + 2ab + b^2, \qquad (a-b)^2 = a^2 - 2ab + b^2,$$
$$(a+b)(a-b) = a^2 - b^2.$$

Write a proof.

Solution:

Expand $(a+b)(a+b)$: $a^2+ab+ba+b^2 = a^2+2ab+b^2$. Replacing b by $-b$ gives the second identity. Finally $(a+b)(a-b) = a^2 - ab + ab - b^2 = a^2 - b^2$.

Example 8.5

Expand $(x+3)^2$, $(3x-4)^2$, $(x-3)(x+3)$, $(2x+3)(2x-3)$, $(4-3x)(3x+4)$.

Solution:

x^2+6x+9 ; $9x^2-24x+16$; x^2-9 ; $4x^2-9$; $16-9x^2$. In the last product, reorder 3 $x+4$ into $4+3x$.

Example 8.6

Expand and simplify $A = (2x-3)^2 + (x+5)(3-x)$ and $B = 2(x+3) + (2x+3)(2x-3)$.

Solution:

$A = 4x^2 - 12x + 9 - x^2 - 2x + 15 = 3x^2 - 14x + 24$. $B = 2x + 6 + 4x^2 - 9 = 4x^2 + 2x - 3$.

IV Special products: factorising

Method

Spot the two squares; for three-term expressions, check the cross term 2 ab. For a difference of two squares, use $a^2 - b^2 = (a - b)(a + b)$.

Example 8.7

Factorise:

$$A = x^2 - 2x + 1, \quad B = 25 + 16x^2 - 40x, \quad C = 4x^2 + 12x + 9,$$
$$D = 1 - 49x^2, \quad E = 9x^2 - 4, \quad F = (2x + 3)^2 - 64.$$

Solution:

$A = (x - 1)^2$. $B = (4x - 5)^2$. $C = (2x + 3)^2$.
$D = (1 - 7x)(1 + 7x)$. $E = (3x - 2)(3x + 2)$.
$F = [(2x + 3) - 8][(2x + 3) + 8] = (2x - 5)(2x + 11)$. For the first three, check the middle term; for F, treat 2 x + 3 as a single block.

Example 8.8: Checking an identity

Are the eqs ($a + b)^2 = a^2 + b^2$ & $a^2 + b^2 = (a - b)(a + b)$ true for all real numbers?

Solution:

No: for $a = b$ = 1, the first would give 4 = 2 and the second 2 = 0. The missing term in the first is 2 ab ; the product of the second is $a^2 - b^2$.

Fluency · 5 to 10 minutes

Calculate 101^2 mentally; give a direction vector of 2 $x + 3y$ = 1; determine [$-1; 3] \cap [2; 5]$.

Answers: 10201 ; $(-3; 2)$; $[2; 3]$.

V Integer powers (positive and negative)

Definition and rules

For $n \geq 1$, a^n is the product of n factors equal to a. For $a \neq 0$, $a^0 = 1$ and $a^{-n} = 1/a^n$. In this lesson, 0^0 is not defined. For $a, b \neq 0$ and integers m, n :

$$a^m a^n = a^{m+n}, \quad \frac{a^m}{a^n} = a^{m-n}, \quad (a^m)^n = a^{mn},$$

$$(ab)^n = a^n b^n, \qquad \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}.$$

For strictly positive exponents, some bases may be zero if the expressions remain defined.

Example 8.9

Simplify $A = 2^4 \times 2^3$, $B = (-5)^{12}/(-5)^3$, $C = (7^3)^6 \times 7^{-5}$, $D = 7^{24} \times 3^2 \times 2^3$. Also calculate $(-3)^5$, 5^1 and 99999^0.

Solution:

$A = 2^7 = 128$. $B = (-5)^9 = -1953125$. $C = 7^{18-5} = 7^{13}$. $D = 72 \times 7^{24}$: we cannot add the exponents of different bases. The last three values are -243, 5, 1.

Watch out. The definition as a repeated product does not apply to a negative exponent. $-3^2 = -9$, while $(-3)^2 = 9$.

Example 8.10

Mentally solve $2^n > 100$ for the smallest natural number n. Write a program that finds it.

Solution:

$2^6 = 64 \leq 100 < 128 = 2^7$, therefore $n = 7$. Initialise $n = 0$, $p = 1$; while $p \leq 100$, replace p by $2p$ and n by $n + 1$. The program is given at the end of the unit.

VI Fractions and fractional expressions

Rules and conditions

For non-zero denominators:

$$\frac{a}{c} \pm \frac{b}{c} = \frac{a \pm b}{c}, \quad \frac{a}{c} \pm \frac{b}{d} = \frac{ad \pm bc}{cd}, \quad \frac{a}{c} \times \frac{b}{d} = \frac{ab}{cd}.$$

A fraction c/d has an inverse only if $c \neq 0$ and $d \neq 0$. Then

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc} \quad (b, c, d \neq 0).$$

Example 8.11

Calculate and write in lowest terms:

$$A = \left(\frac{2}{3} + \frac{1}{6}\right) \times \frac{4}{5}, \quad B = \frac{1}{2} + \frac{3}{4} \times \frac{5}{6}, \quad C = \frac{4}{5} \div \frac{2}{3} \times \frac{7}{10},$$

$$D = \frac{\left(\frac{2}{5} + \frac{1}{3}\right) \times \frac{3}{4}}{\frac{1}{2} - \frac{1}{6}}.$$

Solution:

$A = \frac{5}{6} \times \frac{4}{5} = \frac{2}{3}$. $B = \frac{1}{2} + \frac{5}{8} = \frac{9}{8}$. $C = \frac{4}{5} \times \frac{3}{2} \times \frac{7}{10} = \frac{21}{25}$. $D = \frac{11}{15} \times \frac{3}{4} \div \frac{1}{3} = \frac{11}{20} \times 3 = \frac{33}{20}$. In C, multiplication and division are performed from left to right.

Example 8.12

Simplify $\dfrac{x^2 - 1}{x - 1}$ stating the domain. Can we then take $x = 1$?

Solution:

For $x \neq 1$, $\frac{(x-1)(x+1)}{x-1} = x + 1$. The original expression remains undefined at 1; simplifying does not remove the restriction.

VII Square roots and conditions of validity

Definition and properties

For $a \geq 0$, $\sqrt{a}$ is the unique number, positive or zero, whose square is a. Thus $(\sqrt{a})^2 = a$. For every $x \in \mathbb{R}$, $\sqrt{x^2} = |x|$. For $a, b \geq 0$, $\sqrt{ab} = \sqrt{a}\sqrt{b}$. If $b > 0$, $\sqrt{a/b} = \sqrt{a}/\sqrt{b}$.

Write a proof.

Solution:

The number $\sqrt{a}\sqrt{b}$ is positive or zero and its square is ab ; by uniqueness of the positive square root, it equals $\sqrt{ab}$. The same argument applied to the quotient establishes the second formula, for $b > 0$. In the way $|x| \geq 0$ and $|x|^2 = x^2$, therefore $\sqrt{x^2} = |x|$.

Example 8.13

Simplify $\sqrt{18}$; $\sqrt{2}\sqrt{8}$; $\sqrt{7/25}$; $\sqrt{27/75}$; $\sqrt{(-5)^2}$.

Solution:

$3\sqrt{2}$; 4 ; $\sqrt{7}/5$; $\sqrt{9/25} = 3/5$; 5.

Watch out. There is no general rule $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$. With $a = 4, b = 9$, we would get $\sqrt{13} = 5$. Saying that these expressions are always different would also be false, for example if $a = 0$.

Example 8.14

Simplify $\sqrt{18} - \sqrt{50} + \sqrt{8}$. Bound $\sqrt{20}$ between two consecutive integers.

Solution:

$6\sqrt{2} - 5\sqrt{2} + 2\sqrt{2} = 3\sqrt{2}$. Since $16 < 20 < 25$, $4 < \sqrt{20} < 5$.

VIII First-degree equations

Definition

Solving an equation means finding all the values of the unknown that satisfy the equality. Two equations are equivalent if they have the same solutions. Adding the same expression or multiplying by the same non-zero number preserves equivalence.

Method

Expand if necessary, group the terms in x, then isolate x. If its coefficient is zero, examine the remaining equality: it may be always true or impossible. Check in the original equation.

Example 8.15

Solve:

$$a)\ 5x-3=2x+9, \qquad b)\ 3(2x-1)=2(x+4)+7,$$
$$c)\ -2(x+3)=4x-(x-1), \qquad d)\ \frac{x+2}{3}=x-1,$$
$$e)\ \frac{-5x+5}{8}=-\frac{3}{4}, \qquad f)\ \frac{-3x-1}{9}=-\frac{7}{4}.$$

Solution:

a) $3x = 12$, therefore $x = 4$. b) $6x-3=2x+15$, so $x = 9/2$. c) $-2x-6=3x+1$, therefore $x=-7/5$. d) $x+2=3x-3$, so $x = 5/2$. e) $-5x+5=-6$, so $x = 11/5$. f) $4(-3x-1)=-63$, therefore $-12x=-59$ and $x = 59/12$.

Example 8.16

Solve $2(x+1)=2x+2$, then $2(x+1)=2x+3$.

Solution:

The first reduces to $2 = 2$: $S=\mathbb{R}$. The second gives $2 = 3$: $S=\varnothing$. We do not divide by a zero coefficient.

IX Zero product and equations with a quotient

Properties

$AB = 0 \iff A = 0$ or $B = 0$. For $B \neq 0$, $A/B = 0 \iff A = 0$. To solve $A(x)/B(x) = k$, first require $B(x) \neq 0$, then solve $A(x) = kB(x)$ and exclude the forbidden values.

Example 8.17

Solve $(4x + 6)(3 - 7x) = 0$; $3x(-7x + 5) = 0$; $\dfrac{2x + 1}{3x - 5} = 0$.

Solution:

Solutions: $\{-3/2; 3/7\}$; $\{0; 5/7\}$; $\{-1/2\}$, with forbidden value $5/3$ in the last case.

Example 8.18

Factorise, then solve $(1 - 6x)^2 - (1 - 6x)(2 + 5x) = 0$ and $(4x - 1)(x + 6) + (4x - 1) = 0$.

Solution:

First: $(6x - 1)(11x + 1) = 0$, therefore $S = \{1/6; -1/11\}$. Second: $(4x - 1)(x + 7) = 0$, therefore $S = \{1/4; -7\}$.

Example 8.19

Solve $\dfrac{x + 1}{x - 2} = 3$, then $\dfrac{x^2 - 4}{x - 2} = 4$.

Solution:

In both cases, $x \neq 2$. First: $x + 1 = 3x - 6$, so $x = 7/2$, allowed. Second: for $x \neq 2$, the quotient is $x + 2$; the equation requires $x = 2$, forbidden. So there is no solution.

X Equations with a square; isolate a variable

Reminder

In $\mathbb{R}$, $x^2 = a$ has two solutions $\pm\sqrt{a}$ if $a > 0$, one solution 0 if $a = 0$, and none if $a < 0$.

Example 8.20

Solve $x^2 = 0$; $x^2 = -8$; $x^2 = 7$; $25x^2 = 81$; $(x+2)^2 = 9$.

Solution:

In order: $\{0\}$; $\varnothing$; $\{-\sqrt{7}; \sqrt{7}\}$; $\{-9/5; 9/5\}$; $\{-5; 1\}$. For the last one, $x+2 = -3$ or $x+2 = 3$.

Isolate a variable in a formula

Describe the operations performed on both sides; state the conditions that allow us to divide or take a square root.

Example 8.21

Isolate h in $V = \pi r^2 h$ with $r > 0$; isolate v in $E = \frac{1}{2}mv^2$ with $m > 0$, $E \geq 0$ and $v \geq 0$; isolate x in $y = ax + b$ with $a \neq 0$.

Solution:

$h = V/(\pi r^2)$. Then $v^2 = 2E/m$; since $v \geq 0$, $v = \sqrt{2E/m}$. Finally $x = (y-b)/a$. Without the condition $v \geq 0$, two signs would be possible for $E > 0$.

Fluency · 5 to 10 minutes

Give the domain of $1/(x+3)$; calculate $\sqrt{49}$; the slope between (0; 2) and (3; 8).

Answers: $\mathbb{R} \setminus \{-3\}$; 7 ; 2.

XI Inequalities and comparison

Rules

Inequalities in the same direction can be added: $a \leq b,\ c \leq d \Rightarrow a + c \leq b + d$. Adding the same real number preserves the direction. Multiplying or dividing by a strictly positive real number preserves the direction; by a strictly negative real number, reverses it. Comparing A and B amounts to studying the sign of $A - B$. If $A, B > 0$, we can also compare A/B to 1.

Example 8.22

1. If $x \leq -4$, what about $3x - 4$? 2. If $x > -6$, what about $2x - 3$? 3. If $7 \leq x \leq 16$ and $3 \leq y \leq 12$, bound $8x + 7y$. 4. If $-10 \leq x \leq -6$, bound $2x - 6$.

Solution:

1. $3x - 4 \leq -16$. 2. $2x - 3 > -15$. 3. $56 \leq 8x \leq 128$ and $21 \leq 7y \leq 84$, hence $77 \leq 8x + 7y \leq 212$. The bounds can be reached. 4. $-26 \leq 2x - 6 \leq -18$.

Example 8.23

Solve $x + 1 \geq x - 7$; $2x - 11 \leq 4x - 5$; $6x + 19 \leq 2x - 3$.

Solution:

1. $2x \geq -8$, i.e. $[-4; +\infty[$. 2. $-6 \leq 2x$, i.e. $[-3; +\infty[$. 3. $4x \leq -22$, i.e. $]-\infty; -11/2]$.

XII Sign of an affine expression and of a product

Property

For $a \neq 0$, $ax + b$ is zero at $x_0 = -b/a$; its sign on the right is that of a, the opposite on the left. If a = 0, the sign is that of the constant b. For a product, multiply the signs of factors.

Example 8.24

Draw the sign tables of 3 $-x$, $2x - 6$, then $P(x) = (2x - 6)(x + 1)$.

Solution:

Both affine expressions are zero at 3: 3 $-x$ is positive on the left, negative on the right; $2x - 6$ a the opposite signs.

x	$-\infty$		-1		3		$+\infty$
$2x - 6$		$-$	$-$	$-$	0	$+$	
$x + 1$		$-$	0	$+$	$+$	$+$	
$P(x)$		$+$	0	$-$	0	$+$	

Example 8.25

Solve (7 + 3 $x)(2x + 6) > 0$.

Solution:

The zeros are $-7/3$ and -3, ordered as $-3 < -7/3$. The product is positive outside these roots and negative between them. The inequality is strict: $S =]-\infty; -3[\cup]-7/3; +\infty[$.

XIII Sign of a quotient and algorithmic summary

Method

Find the forbidden values, then the zeros of the numerator. Order all critical values and apply the sign rule. On the quotient row, a double bar indicates a forbidden value.

Example 8.26

Draw the sign table of $Q(x) = \dfrac{-4x+6}{x-3}$, then solve $\dfrac{2x-1}{-4x+3} \leq 0$.

Solution:

For Q, zero 3 /2, forbidden value 3. Signs: negative on] $-\infty; 3/2[$, zero at 3 /2, positive on]3 /2; 3[, not defined at 3, negative on]3; + ∞[. For the second expression, zero 1 /2, forbidden value 3/4. The signs are $-, +, -$ on the three intervals. Thus $S =]-\infty; 1/2] \cup]3/4; +\infty[$.

```
def premiere_puissance(seuil):
    # seuil reel >= 1
    n, p = 0, 1
    while p <= seuil:
        n = n+1
        p = 2*p
    return n, p
```

Example 8.27

What does `premiere_puissance(100)` return? What link holds between `p` and `n` at every stage? How then test the sign of a number in Python?

Solution:

The result is (7 , 128). The invariant is $p = 2^n$, true at initialisation and then preserved. For the sign, test first `x > 0`, then `x < 0`, otherwise x = 0. Do not replace the loop's non-strict inequality with a strict one without changing the problem posed.